



Computing Infrastructures

July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

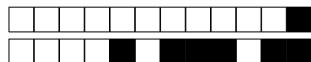
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A False

☐ B True

Question 2 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

☐ A False

☐ B True

Question 3 Docker and Kubernetes are two technology examples supporting para-virtualization.

☐ A False

☐ B True

Question 4 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A False

☐ B True

Question 5 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A True

☐ B False

Question 6 Replicating data across availability zones helps ensure data durability in case of local failures.

☐ A True

☐ B False

Question 7 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A True

☐ B False

Question 8 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A True

☐ B False

Question 9 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A True

☐ B False

Question 10 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.



Question 13

Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 600$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



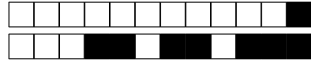
+1/5/56+

Question 15

Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 5req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 55% (assume that the servers of the same type evenly distribute their load).

Question 16

Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?



Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

[illegible]





+1/9/52+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona): SOL

True/False Questions

- Question 01 : ☒ A ☐ B
Question 02 : ☒ A ☐ B
Question 03 : ☒ A ☐ B
Question 04 : ☐ A ☒ B
Question 05 : ☒ A ☐ B
Question 06 : ☒ A ☐ B
Question 07 : ☒ A ☐ B
Question 08 : ☐ A ☒ B
Question 09 : ☒ A ☐ B
Question 10 : ☒ A ☐ B

Exercises

Question 11 : 32 SERVERS

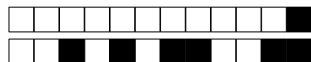
Question 12 : 0,9957 = A_{vA} A_{SYS} = 0,9584

Question 13 : 15.1 $\Rightarrow$ 14 DISKS

Question 14 : D_{WEB} = 2,25 sec D_{APP} = 1,5 sec D_{DBMS} = 1,8 sec

Question 15 : N_{WEB} = 21 N_{APP} = 14 N_{DBMS} = 17

Question 16 : X_{MAX} = 9,3 REQ/SEC



Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

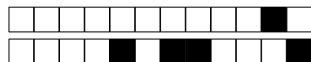
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A True

☐ B False

Question 2 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A True

☐ B False

Question 3 Docker and Kubernetes are two technology examples supporting para-virtualization.

☐ A True

☐ B False

Question 4 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A True

☐ B False

Question 5 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A True

☐ B False

Question 6 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A False

☐ B True

Question 7 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A True

☐ B False

Question 8 Replicating data across availability zones helps ensure data durability in case of local failures.

☐ A True

☐ B False

Question 9 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

☐ A True

☐ B False

Question 10 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A data center hosts a critical service that depends on a system with three components:

- Component A is a central control unit (such as a network controller or orchestrator) essential for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (like application servers) configured so that only one needs to be operational for the system to work correctly.

We know that the reliability of A after 3 years is 0.6, while its MTTR (Mean Time to Repair) is 21 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.65 and 0.8. Compute the availability of component A and compute the overall system availability.



Question 13

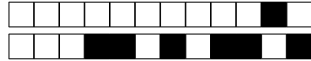
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 7 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 6 years

Question 14

An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 600$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



+2/6/45+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

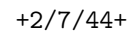
⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the entire width of the page. There are no margins, text, or other markings present.





+2/9/42+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

- Question 01 : ☒ A ☐ B
Question 02 : ☒ A ☐ B
Question 03 : ☐ A ☒ B
Question 04 : ☒ A ☐ B
Question 05 : ☒ A ☐ B
Question 06 : ☒ A ☐ B
Question 07 : ☐ A ☒ B
Question 08 : ☒ A ☐ B
Question 09 : ☐ A ☒ B
Question 10 : ☒ A ☐ B

Exercises

Question 11 : 32 SERVERS

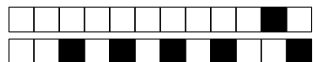
Question 12 : $A_A = 0,8803$ $A_{SYS} = 0,8210$

Question 13 : 11,5 $\Rightarrow$ 10 DISKS

Question 14 : $D_{WEB} = 2,25 \text{ sec}$ $D_{APP} = 1,5 \text{ sec}$ $D_{DBMS} = 1,8 \text{ sec}$

Question 15 : $N_{WEB} = 21$ $N_{APP} = 14$ $N_{DBMS} = 17$

Question 16 : $X_{MAX} = 8,3 \text{ REQ/SEC}$



Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

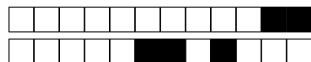
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

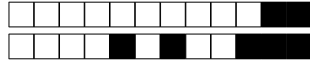
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is **ONLY** the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A False

☐ B True

Question 2 Replicating data across availability zones helps ensure data durability in case of local failures.

☐ A True

☐ B False

Question 3 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A True

☐ B False

Question 4 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

☐ A True

☐ B False

Question 5 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A True

☐ B False

Question 6 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A True

☐ B False

Question 7 Docker and Kubernetes are two technology examples supporting para-virtualization.

☐ A False

☐ B True

Question 8 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A False

☐ B True

Question 9 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A True

☐ B False

Question 10 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.



Question 13

Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 300$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What are the service demand at the three servers? What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



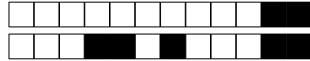
+3/5/36+

Question 15

Given the system described at Question 14, you expect the incoming load to grow to $\lambda = 3req/s$. Determine the minimum number of servers at each physical layer in a way that the utilization of each server is at most 70% (assume that the servers of the same type evenly distribute their load).

Question 16

Assume you upgraded the system according to your answer to Question 15. What is the maximum throughput that the new system can achieve?



+3/6/35+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

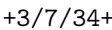
⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

First Name (CAPITAL LETTERS):

Last Name (CAPITAL LETTERS):

Student ID (Codice Persona):

Question 17

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?





+3/9/32+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

- Question 01 : ☐ A ☒ B
Question 02 : ☒ A ☐ B
Question 03 : ☒ A ☐ B
Question 04 : ☐ A ☒ B
Question 05 : ☐ A ☒ B
Question 06 : ☒ A ☐ B
Question 07 : ☒ A ☐ B
Question 08 : ☐ A ☒ B
Question 09 : ☒ A ☐ B
Question 10 : ☐ A ☒ B

Exercises

Question 11 : 32 SERVERS

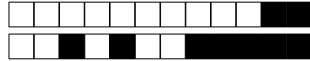
Question 12 : $A_A = 0,9957$ $A_{SYS} = 0,9584$

Question 13 : 15.1 = 14 DISKS

Question 14 : $D_{WEB} = 4.5 \text{ sec}$ $D_{APP} = 3 \text{ sec}$ $D_{DBMS} = 3.6 \text{ sec}$

Question 15 : $N_{WEB} = 20$ $N_{APP} = 13$ $N_{DBMS} = 16$

Question 16 : $X_{MAX} = 4,3 \text{ REQ/sec}$



Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

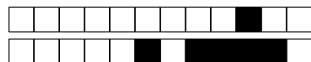
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



Computing Infrastructures
July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

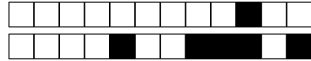
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the Open Questions should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A False

☐ B True

Question 2 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A True

☐ B False

Question 3 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A False

☐ B True

Question 4 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A False

☐ B True

Question 5 Docker and Kubernetes are two technology examples supporting para-virtualization.

☐ A False

☐ B True

Question 6 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

☐ A False

☐ B True

Question 7 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A True

☐ B False

Question 8 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A True

☐ B False

Question 9 Replicating data across availability zones helps ensure data durability in case of local failures.

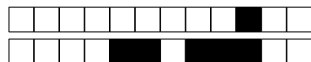
☐ A True

☐ B False

Question 10 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A data center hosts a critical service that depends on a system with three components:

- Component A is a central control unit (such as a network controller or orchestrator) essential for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (like application servers) configured so that only one needs to be operational for the system to work correctly.

We know that the reliability of A after 3 years is 0.6, while its MTTR (Mean Time to Repair) is 21 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.65 and 0.8. Compute the availability of component A and compute the overall system availability.



Question 13

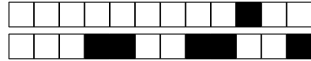
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$ you get the following data:

- Number of completions $C = 300$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What are the service demand at the three servers? What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



+4/6/25+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

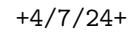
⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

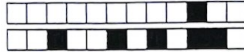
Student ID (Codice Persona):

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

This image shows a full page of white paper with horizontal dotted lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.



●



+4/9/22+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

- Question 01 : ☒ A ☐ B
Question 02 : ☐ A ☒ B
Question 03 : ☐ A ☒ B
Question 04 : ☐ A ☒ B
Question 05 : ☒ A ☐ B
Question 06 : ☒ A ☐ B
Question 07 : ☒ A ☐ B
Question 08 : ☒ A ☐ B
Question 09 : ☒ A ☐ B
Question 10 : ☒ A ☐ B

Exercises

Question 11 : 32 SERVERS

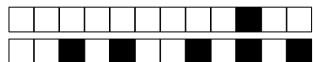
Question 12 : $A_A = 0,9203$ $A_{sys} = 0,9210$

Question 13 : 15.1 $\Rightarrow$ 14 DISKS

Question 14 : $D_{WEB} = 4.5 \text{ sec}$ $D_{APP} = 3 \text{ sec}$ $D_{DBMS} = 3.6 \text{ sec}$

Question 15 : $N_{WEB} = 20$ $N_{APP} = 13$ $N_{DBMS} = 16$

Question 16 : $X_{MAX} = 4.3 \frac{\text{Req}}{\text{sec}}$



+4/10/21+

Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

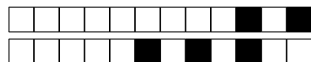
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



+5/1/20+

Computing Infrastructures

July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

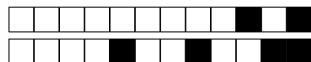
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A True

☐ B False

Question 2 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A True

☐ B False

Question 3 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A False

☐ B True

Question 4 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A True

☐ B False

Question 5 Replicating data across availability zones helps ensure data durability in case of local failures.

☐ A True

☐ B False

Question 6 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A False

☐ B True

Question 7 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A True

☐ B False

Question 8 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A False

☐ B True

Question 9 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

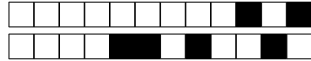
☐ A False

☐ B True

Question 10 Docker and Kubernetes are two technology examples supporting para-virtualization.

☐ A True

☐ B False



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.



Question 13

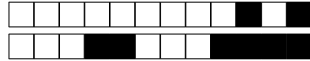
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 420 days and a Mean Time To Repair (MTTR) of 4 days. What is the maximum number of disks that can be included while ensuring that the overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet-based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$, you get the following data:

- Number of completions $C = 300$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



+5/6/15+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

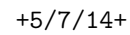
⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

[illegible]



●



+5/9/12+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

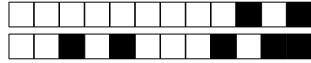
Student ID (Codice Persona):

True/False Questions

- Question 01 : ☒ A ☐ B
Question 02 : ☒ A ☐ B
Question 03 : ☒ A ☐ B
Question 04 : ☒ A ☐ B
Question 05 : ☒ A ☐ B
Question 06 : ☐ A ☒ B
Question 07 : ☒ A ☐ B
Question 08 : ☒ A ☐ B
Question 09 : ☒ A ☐ B
Question 10 : ☐ A ☒ B

Exercises

- Question 11 : 32 SERVERS
- Question 12 : $A_A = 0,9957$ $A_{SYS} = 0,9584$
- Question 13 : 15.1 $\rightarrow$ 14 DISKS
- Question 14 : $D_{WEB} = 4,5 \text{ sec}$ $D_{APP} = 3 \text{ sec}$ $D_{DBMS} = 3,65 \text{ sec}$
- Question 15 : $N_{WEB} = 25$ $N_{APP} = 17$ $N_{DBMS} = 20$
- Question 16 : $X_{MAX} = 5,5 \frac{\text{Req}}{\text{SEC}}$



Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.



+6/1/10+

Computing Infrastructures

July 4, 2025

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

Write the LAST and FIRST name in CAPITAL LETTER, and in this order, in all places where requested. **Where it is requested only the STUDENT ID (Codice Persona), do not write your name.**

Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is ONLY the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.

**True false questions**

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Cloud-native applications are designed to run in the cloud and often use microservices and containers.

☐ A False

☐ B True

Question 2 Replicating data across availability zones helps ensure data durability in case of local failures.

☐ A True

☐ B False

Question 3 3-tier datacenter networks consist of three distinct layers named core, aggregation, and access.

☐ A False

☐ B True

Question 4 All hardware accelerators used in the datacenter context use seamless programming interfaces and software frameworks, and can be easily interchanged.

☐ A False

☐ B True

Question 5 Data centers using HDDs typically ignore redundancy features, assuming disk failures are very rare.

☐ A True

☐ B False

Question 6 Warehouse-scale computers are single servers that are designed to be used in large-scale data centers.

☐ A True

☐ B False

Question 7 The average seek time of an HDD can be estimated as one-third of the time it takes the actuator arm to move from the innermost to the outermost track.

☐ A False

☐ B True

Question 8 Docker and Kubernetes are two technology examples supporting paravirtualization.

☐ A False

☐ B True

Question 9 Open-loop cooling exploits favourable outdoor conditions to reduce mechanical cooling load.

☐ A True

☐ B False

Question 10 East-west traffic refers to network traffic that travels between servers within the same datacenter.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?

Question 12

A datacenter hosts a critical service that relies on a system composed of three components:

- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.



+6/4/7+

Question 13

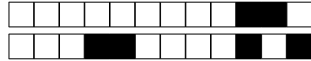
Your task is to design a storage system using a RAID 1+0 configuration, consisting of HDDs with a capacity of 24 GB each. Each disk has a Mean Time To Failure (MTTF) of 520 days and a Mean Time To Repair (MTTR) of 7 days. What is the maximum number of disks that can be included while ensuring that the MTTF for overall storage ($MTTF_{RAID1+0}$) is at least 8 years

Question 14

An Internet-based system includes initially a Web server, an application server, and a DBMS. After monitoring the system for $T = 1800s$, you get the following data:

- Number of completions $C = 300$
- Web server utilization $U_{Web} = 75\%$
- Application server utilization $U_{App} = 50\%$
- DBMS utilization $U_{DBMS} = 60\%$

What is the service demand for the three servers (i.e. D_{Web} , D_{App} , D_{DBMS})?



+6/6/5+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

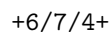
⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

Question 18

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ How does the geographical distribution of datacenters affect aspects such as latency, reliability, and environmental sustainability?

[illegible]

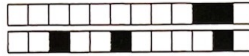


●

Answer Sheets (Page 2)

⇒ Compare the main storage technologies used in datacenters, such as HDDs, SSDs, and tapes. In which scenarios is each one preferable? Discuss advantages, disadvantages, and selection criteria in terms of performance, cost, capacity, and reliability.





+6/9/2+

Computing Infrastructures - July 4, 2025

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

- Question 01 : ☐ A ☒ B
Question 02 : ☒ A ☐ B
Question 03 : ☐ A ☒ B
Question 04 : ☒ A ☐ B
Question 05 : ☐ A ☒ B
Question 06 : ☐ A ☒ B
Question 07 : ☐ A ☒ B
Question 08 : ☒ A ☐ B
Question 09 : ☒ A ☐ B
Question 10 : ☐ A ☒ B

Exercises

Question 11 : 32 SERVERS

Question 12 : $A_A = 0,9957$ $A_{ys} = 0,9584$

Question 13 : 13.2 $\Rightarrow$ 12 DISKS

Question 14 : $D_{WER} = 4.5 \text{ sec}$ $D_{AP} = 3 \text{ sec}$ $D_{DBMS} = 3.6 \text{ sec}$

Question 15 : $N_{WRB} = 25$ $N_{AP} = 17$ $N_{DBMS} = 20$

Question 16 : $X_{MAX} = 5,5 \frac{\text{REQ}}{\text{SEC}}$



Computing Infrastructures
July 4, 2025

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

⇒ **The remaining part of this page has been intentionally left blank** ⇐

If needed, you can use this page for notes. Any answer written here will be ignored.